# MALLIKARJUNA THIPPANA, PhD

**Computational Biologist | Cancer Gene Regulation, 3D Genomics & Computational Methods**

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## PROFESSIONAL PROFILE

Computational biologist with a PhD in Computational Systems Biology and research experience in cancer regulatory genomics, MYC/CTCF enhancer control, cancer whole-genome analysis, transcriptomics and integrative multi-omics. At Karolinska Institutet, I analyse TF-associated regulatory landscapes using GRO-seq, RNA-seq, ATAC-seq, CUT&RUN and 4C-seq evidence; lead a 17-donor clinical snRNA-seq analysis; analyse cancer whole-genome data for focal amplifications and ecDNA; and develop reproducible Python/R workflows. During my PhD, I designed two peer-reviewed, open-source gene-prioritisation algorithms.

## CORE EXPERTISE

- **Gene regulation & 3D genomics:** MYC/CTCF enhancer and long-range regulatory analysis; integration of 4C-seq, CTCF binding, enhancer annotations, CUT&RUN and nascent-transcription responses.
- **Epigenomics & transcriptomics:** ATAC-seq, GRO-seq, RNA-seq, ChIP-seq, CUT&RUN; peak and motif analysis, differential accessibility/expression, pathway and TF enrichment.
- **Cancer genomics:** cancer WGS with AmpliconSuite (AmpliconArchitect, AmpliconClassifier) and nf-core/circdna; somatic and copy-number interpretation (GATK, bcftools); TCGA, GEO, COSMIC, cBioPortal.
- **Single-cell:** 10x sc/snRNA-seq with Scanpy/scverse, Seurat and Harmony; multi-donor QC, integration, annotation, differential testing and variability.
- **Methods & statistics:** two original published gene-prioritisation algorithms; WGCNA and network biology; statistical modelling of high-dimensional data.
- **Reproducibility:** Python, R, Bash, Git/GitHub Actions, Nextflow/nf-core, Snakemake, Docker/Apptainer, Linux HPC/SLURM.

## RESEARCH EXPERIENCE

**Postdoctoral Researcher – Cancer Regulatory Genomics & Multi-Omics** **Dec 2024 – Present**

*Department of Oncology-Pathology, Karolinska Institutet, Solna, Sweden | PI: Dr Anita Göndör*

- Contribute to MYC/CTCF cancer-regulatory genomics combining GRO-seq, RNA-seq, ATAC-seq, 4C-seq and experimental evidence to study enhancer activity, long-range regulation and MYC expression plasticity; manuscript under revision at *Nucleic Acids Research*.
- Analyse NUP133 CUT&RUN occupancy relative to CTCF sites and enhancer annotations; TF-motif and CTCF-regulatory enrichment analyses to prioritise hypotheses for experimental tests.
- Integrate ATAC-seq (5,175 differential regions), RNA-seq (305 DEGs) and GRO-seq (1,816 DEGs) for p60AmotL2/iBET151 perturbations; identify 98 shared and 38 chromatin-priming candidates.
- Lead computational analysis of a 17-donor endometrial snRNA-seq clinical cohort (5 controls, 12 PCOS): QC, Harmony integration, cell-type annotation, cell-type-specific differential expression and variability, CTCF-binding enrichment.
- Built and validated an ecDNA workflow (AmpliconSuite, nf-core/circdna) on public WGS of the MYCN-amplified LAN-5 neuroblastoma cell line, reconstructing and classifying focal amplicons: the MYCN amplicon (copy number ≈212) was classified ecDNA-positive, with a 455 kb candidate cycle closed by a junction supported by 241 reads. The pipeline is next being applied to the group's tumour WGS data.

**IoE Postdoctoral Fellow – Computational Drug Discovery** **Jul 2024 – Nov 2024**

*University of Hyderabad, India*

- Applied Python-supported docking, pharmacophore and ADMET analyses to rank candidate anti-cancer tyrosine-kinase inhibitors.

**PhD Researcher – Cancer Systems Biology & Computational Methods** **2019 – 2024**

*Department of Biotechnology & Bioinformatics, University of Hyderabad, India | ICMR-funded*

- Analysed TCGA/GEO cancer cohorts with transcriptomic, genomic-variation, network, pathway and survival evidence to identify molecular regulators and disease-associated genes.
- Conceived, implemented and published two original Python/R gene-prioritisation algorithms (*Theory in Biosciences*, 2024; *Proteins*, 2022) with open-source implementations.

- Applied WGCNA, protein-interaction and regulatory-network models; contributed to cervical and lung cancer studies; supervised five MSc dissertations.

**Junior Research Fellow – Network Biology & Systems Bioinformatics** **2018 – 2019**
*CRRao AIMSCS & School of Physics, University of Hyderabad, India | DST-SERB-funded*

## EDUCATION

**PhD, Biotechnology – Computational Systems Biology** (2018–2024), University of Hyderabad, India. Thesis: *Integrative studies to explore key molecular players involved in cervical squamous cell carcinoma*. Defended 20 May 2024.

**M.Tech, Bioinformatics** (2015–2017), University of Hyderabad, India. GATE 2015 qualified.

## AWARDS, PRESENTATIONS & SERVICE

- **Awards:** ICMR Senior Research Fellowship (2019); DST-SERB International Travel Support and ISCB Travel Fellowship (ISMB/ECCB 2023); NUS–Temasek Foundation Travel Grant (2023); DBT-CTEP Travel Grant (2022).
- **Presentations:** ISMB/ECCB 2023, Lyon (flash talk + poster, iRNA track); GIW XXXI/ISCB-Asia V 2022, Tainan (paper); Systems Approaches to Cancer Biology 2022, Woods Hole (selected poster).
- **Memberships:** ISCB; RNA Society; FABA. **Languages:** English (fluent), Telugu (native), Hindi (basic), Swedish (basic).
- **Open-source software:** github.com/arjunbioinfo — CI-tested, Zenodo-archived implementations of both published methods; ecDNA/WGS and scRNA-seq pipelines.

## PUBLICATIONS

**Peer-reviewed journal articles (8)**

1. Dwivedi A, **Thippana M**, Khammampalli S, Cholleti SN, Vindal V. Unraveling the gender-specific molecular landscape of lung squamous cell carcinoma progression. *J Biomol Struct Dyn*, 2026; 44(3):1348–1361.
2. **Mallikarjuna T**, Thummadi NB, Vindal V, Manimaran P. Prioritizing cervical cancer candidate genes using chaos game and fractal-based time series approach. *Theory in Biosciences*, 2024; 143(3):183–193.
3. **Thippana M**, Dwivedi A, Das A, Palanisamy M, Vindal V. Identification of key molecular players and associated pathways in cervical squamous cell carcinoma progression through network analysis. *Proteins*, 2023; 91(8):1173–1187.
4. Prabakaran DS, Chaturvedi PK, Krishnamoorthy D, Seo YS, **Thippana M**, Park WY. Fused toes homolog, a potential molecular regulator of HPV16 E6 and E7 oncoproteins in cervical cancer. *PLOS ONE*, 2022; 17(4):e0266532.
5. Thummadi NB, **Mallikarjuna T**, Vindal V, Manimaran P. Prioritizing the candidate genes related to cervical cancer using the moment of inertia tensor. *Proteins*, 2022; 90(2):363–371.
6. Allam M, Bhavani AKD, Mudiraj A, Ranjan N, **Thippana M**, Babu PP. Synthesis of pyrazolo[3,4-d]pyrimidin-4(5H)-ones tethered to 1,2,3-triazoles and their evaluation as potential anticancer agents. *Eur J Med Chem*, 2018; 156:43–52.
7. Naveen B, Ommi NB, Mudiraj A, **Thippana M**, Babu PP, Nagarajan R. Total synthesis of Penicinoline E, Marinamide, Methyl Marinamide and their antimalarial activity. *ChemistrySelect*, 2017; 2(11):3256–3261.
8. **Thippanna M**, Subramani PA, Lomada D, Narala VR, Reddy MC. A homology based model and virtual screening of inhibitors for human Geranylgeranyl Transferase 1 (GGTase1). *Bioinformation*, 2013; 9(19):973–977.

**Preprints and manuscripts (3)**

1. Gao C, Martino M, Chaurasiya A, Casagrande FBC, Lim JP, van Ekelenburg YS, Feng W, Liu Y, Scholz B, Mishra P, **Thippana M**, et al. A distal CTCF-binding site drives MYC expression plasticity in a negative feed-forward loop. Under revision at *Nucleic Acids Research*; bioRxiv 2026. doi:10.64898/2026.01.28.702300.
2. Dwivedi A, Zulfia SA, **Thippana M**, Cholleti SN, Vindal V. LCLNCRdb: a comprehensive resource for investigating long non-coding RNAs in lung cancer. bioRxiv, 2025. doi:10.1101/2025.02.14.638263.
3. Dwivedi A, **Thippana M**, Manimaran P, Vindal V. Prioritization of lung cancer candidate genes using moment of inertia tensor analysis. bioRxiv, 2024. doi:10.1101/2024.10.18.619023.

**References** available on request.